

REGINA, SK, Canada

WMO: 715140

Lat: 50.4269N

Lon: 104.6703W

Elev: 577

StdP: 94.58

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.9	-37.5	0.1	-32.7	-34.3	0.2	-29.7	15.6	-10.3	13.7	-10.5	3.7	290	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	30.9	19.2	28.8	18.4	27.1	17.6	21.4	27.5	20.0	26.4	18.8	25.0	5.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	15.1	24.8	17.8	13.7	23.2	16.4	12.5	22.0	65.3	27.3	60.0	26.3	55.8	25.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.5	10.3	-36.7	34.7	2.9	1.9	-38.8	36.0	-40.5	37.2	-42.1	38.2	-44.2	39.6		
(5)			-36.8	24.0	2.9	1.6	-38.9	25.1	-40.6	26.1	-42.2	27.0	-44.2	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-14.5	-13.0	-5.7	3.9	10.9	15.9	18.8	17.9	12.6	4.5	-4.7	-11.9	(6)
(7)		DBStd	13.42	8.42	7.97	7.93	5.89	4.70	3.61	3.14	3.56	4.69	5.18	6.94	7.90	(7)
(8)		HDD10.0	3461	761	643	488	193	47	2	0	0	25	183	441	679	(8)
(9)		HDD18.3	5720	1019	876	746	432	234	91	33	52	178	430	691	938	(9)
(10)		CDD10.0	895	0	0	0	11	73	178	273	244	104	11	0	0	(10)
(11)		CDD18.3	112	0	0	0	0	3	17	48	38	7	0	0	0	(11)
(12)		CDH23.3	1594	0	0	0	7	92	226	560	537	167	6	0	0	(12)
(13)		CDH26.7	475	0	0	0	1	19	56	167	182	50	1	0	0	(13)
(14)	Wind	WSAvg	5.1	5.2	5.0	5.5	5.6	5.5	4.9	4.4	4.4	4.8	5.1	5.1	5.1	(14)
(15)	Precipitation	PrecAvg	395	15	13	18	23	53	75	64	42	34	23	14	18	(15)
(16)		PrecMax	586	40	32	56	63	118	203	182	116	88	107	38	36	(16)
(17)		PrecMin	224	0	1	0	0	9	16	7	4	3	0	0	4	(17)
(18)		PrecStd	83	8	7	12	16	29	46	41	28	20	21	10	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.5	14.4	23.6	29.2	31.5	33.2	33.8	31.1	23.7	14.8	6.0	(19)
(20)			MCWB	1.6	2.1	7.7	11.9	15.2	19.2	21.1	19.6	17.3	13.2	8.2	2.6	(20)
(21)		2%	DB	1.9	2.0	9.7	20.0	26.0	28.0	30.5	30.9	28.0	19.3	9.7	3.1	(21)
(22)			MCWB	0.4	0.6	5.6	10.4	14.1	17.7	20.5	18.9	16.1	11.2	5.5	1.2	(22)
(23)		5%	DB	-0.2	0.2	6.0	17.0	23.2	25.9	28.3	28.5	24.8	15.7	6.3	0.9	(23)
(24)			MCWB	-1.3	-0.7	3.3	8.8	12.6	16.8	19.5	18.2	14.7	9.3	3.3	-0.4	(24)
(25)		10%	DB	-2.9	-1.8	3.3	13.9	20.7	23.8	26.5	26.4	21.9	12.8	3.9	-1.5	(25)
(26)			MCWB	-3.6	-2.6	1.6	7.5	11.5	15.6	18.7	17.3	13.8	7.9	1.8	-2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.5	8.4	13.0	16.8	21.7	24.4	22.3	18.9	14.1	8.6	2.9	(27)
(28)			MCDWB	3.3	4.1	13.3	22.0	24.6	28.6	29.7	28.7	27.3	21.7	14.4	5.4	(28)
(29)		2%	WB	0.5	0.7	5.6	11.2	14.9	19.4	22.2	20.6	17.1	11.9	5.7	1.2	(29)
(30)			MCDWB	1.8	1.7	9.5	18.7	23.3	25.3	28.0	27.7	25.0	18.1	9.3	2.9	(30)
(31)		5%	WB	-1.3	-0.7	3.4	9.5	13.7	17.8	20.6	19.3	15.6	9.9	3.5	-0.4	(31)
(32)			MCDWB	-0.2	0.2	5.8	15.8	21.3	23.6	26.3	26.5	23.3	14.9	5.9	0.9	(32)
(33)		10%	WB	-3.6	-2.7	1.7	7.8	12.4	16.5	19.4	18.2	14.2	8.3	1.9	-2.4	(33)
(34)			MCDWB	-2.8	-1.8	3.2	13.3	19.3	22.1	25.2	24.7	20.7	12.4	3.8	-1.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.0	9.7	12.2	13.8	12.1	13.1	14.0	13.9	12.0	10.0	9.3	(35)
(36)			MCDBR	11.3	10.7	12.2	18.0	18.2	15.8	15.9	17.7	19.2	17.7	13.9	11.6	(36)
(37)			MCWBR	10.3	9.4	8.5	10.0	9.1	7.7	8.1	8.0	9.4	10.6	9.9	9.9	(37)
(38)		5% WB	MCDBR	11.2	10.3	11.6	16.8	15.4	13.0	13.9	15.1	16.6	16.3	12.9	11.2	(38)
(39)			MCWBR	10.3	9.2	8.4	9.9	8.2	7.2	7.9	7.8	8.6	10.0	9.3	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.312	0.272	0.327	0.345	0.350	0.359	0.368	0.325	0.279	0.249	0.239	(40)	
(41)		taud	2.381	2.150	2.507	2.395	2.395	2.424	2.414	2.369	2.465	2.564	2.516	2.404	(41)	
(42)		Ebn at Noon	830	806	950	918	911	906	892	864	872	868	819	774	(42)	
(43)		Edh at Noon	60	101	85	107	112	110	110	109	89	66	53	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.27	2.33	3.68	4.82	5.82	6.08	6.57	5.20	3.88	2.32	1.27	0.90	(44)	
(45)		RadStd	0.16	0.21	0.31	0.38	0.49	0.40	0.40	0.35	0.34	0.29	0.18	0.11	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page